
Statistical Inference for Explainable Boosting Regression

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Abstract

Explainable boosting machines (EBMs) are popular “glass-box” models that maintain accuracy by boosting univariate trees and achieve explainability through clear visualizations of each feature’s effect. However, unlike linear model coefficients, univariate EBM trees lack statistical inference, making it hard to know which features truly matter. We address this gap using recent advances in statistical inference for gradient boosting, providing methods for statistical inference as well as end-to-end theoretical guarantees. Using a moving average instead of a sum of boosted trees (Boulevard regularization) allows the boosting process to converge to a feature-wise kernel ridge regression. This produces asymptotically normal predictions that are only quadratically worse than the minimax rate for nonparametric Lipschitz regression. We construct prediction intervals for the response and confidence intervals for each univariate predictor, enabling rigorous hypothesis tests for feature importance and further explainability within EBMs.

1 Introduction

Modern machine learning methods have proved highly successful but have been critiqued for their lack of transparency, often described as resulting in “black box” models. This, in turn, has led to methods either designed to elucidate their behavior (LIME (Ribeiro et al., 2016), SHAP (Lundberg and Lee, 2017), ALE (Apley and Zhu, 2020) and Partial Dependence (Friedman, 2001) Plots) or to produce inherently transparent models that can be directly inspected by humans

(Rudin, 2019). In light of this, Explainable Boosting Machines (EBM) Lou et al. (2012) have emerged as a popular ‘glass-box’ model to strike the balance between both. In essence, an EBM is built to learn Generalized Additive Model (GAM) Hastie and Tibshirani (1986), where a signal $\mathbf{f}$ is assumed to be additive, i.e. $\mathbf{f} = \sum_{k=1}^p \mathbf{f}^{(k)}$, where each $\mathbf{f}^{(k)}$ is assumed to be a univariate function of the k -th feature. This design allows the measurement of the marginal contribution of each feature like linear models, and where a justification for all the interpretation methods above is that they return the “correct” value for additive models. While EBMs impose strong structural constraints on the resulting prediction, their ability to reproduce nonlinearities, along with appropriate choices of additive structure, provides significant performance, often comparable to SoTA models such as XGBoost Chen and Guestrin (2016), CatBoost Prokhorenkova et al. (2018), LightGBM Ke et al. (2017).

While EBMs Lou et al. (2012) provide a clear graphical representation for the effects of each feature, these are still point estimates. The central question we ask for uncertainty quantification can be phrased as this: *Given a model design, if we collect a new dataset and retrain the model, how different would our new prediction be?* In response to this question, vanilla EBMs Lou et al. (2012) proposed variance estimates via bootstrapping, providing coarse uncertainty quantification, thought at the cost of more complex computation.

In contrast to EBMs, spline-based GAMs enjoy a richer asymptotic theory. Wahba (1983) established the asymptotic distribution of smoothing spline estimators via a Bayesian posterior covariance argument, Silverman (1985) developed the equivalent kernel representation and convergence theory, and Nychka (1988) analyzed the frequentist coverage of Bayesian confidence intervals for smoothing splines. Built on these developments, Wood (2006) investigated the coverage of penalized regression spline intervals in the GAM setting, while Yoshida and Naito (2012) provided a full asymptotic distribution theory for penalized spline estimators, establishing asymptotic normality for each additive component.

Despite this development Lou et al. (2012) argued for using boosted trees to represent additive components due to their adaptive smoothing behavior, providing performance advantages over splines, resulting in the EBM model. Statistical theory for boosting methods is limited to small numbers of boosting rounds in Bühlmann (2002) and extended to structurally constrained models including GAMs in Zhou and Hooker (2022). Here we employ the Boulevard regularization method (Zhou and Hooker, 2019), replacing sums of trees with averages in the training process, which we adapt to the additive model structure of EBMs. Boulevard, particularly in conjunction with subsampling, naturally avoids overfitting; this both downweights the influence of initial trees and allows us to study the model in the limit of infinite boosting rounds in which we can produce theoretical results without the need to select or constrain the number of rounds. In particular, our work makes the following contributions

- **Asymptotic Theory.** We establish a central limit theorem for EBM predictors, showing that the Boulevard-regularized ensemble converges to a feature-wise kernel ridge regression. The resulting predictions are asymptotically normal and achieve rates only quadratically worse than the minimax rate for Lipschitz regression.
- **Frequentist Inference.** By leveraging our proposed CLT for baseline vector β and feature-wise predictions $\hat{\mathbf{f}}^{(k)}$, we extend the heuristic bootstrap to a set of inference toolkits: Formal confidence intervals, prediction intervals and reproduction intervals Zhou and Hooker (2019). This also allows a direct test of variable importance following the methods of Mentch and Hooker (2015). We round out those propositions by numerical experiments to back up the validity of our algorithms.
- **Computational Efficiency.** Unlike prior inferential approaches for random forests and boosting, we exploit the Nyström approximation of Musco and Musco (2017), allowing linear time inferences.

Interaction models?

exploitation of basis representation?

need to sketch out results and simulations

2 Setup

In this section, we introduce the problem setup with specified notations, as well as the assumptions this study build on.

Our goal is to learn a signal $\mathbf{f}$ that is additive with the form as this.

Assumption 2.1 (Additive Lipschitz Nonparametric Regression). Let the response satisfy

$$\mathbf{y} = \beta + \sum_{k=1}^p \mathbf{f}^{(k)}(\mathbf{x}^{(k)}) + \epsilon := \mathbf{f}(\mathbf{x}) + \epsilon,$$

where $\mathbf{y}, \beta \in \mathbb{R}$, $\mathbf{x} = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(p)})^\top \in \mathbb{R}^p$, and ϵ are i.i.d. sub-Gaussian errors with mean zero and variance $\sigma^2 < \infty$. Assume the covariates have a bounded density on their support: $\exists 0 < c_1 \leq \mu(\mathbf{x}) \leq c_2 < \infty$. Each component function $\mathbf{f}^{(k)} : \mathbb{R} \rightarrow \mathbb{R}$ is mean-zero with respect to the marginal of $\mathbf{x}^{(k)}$, and is L -Lipschitz continuous. The intercept vector β is defined as the unconditional mean of the response,

$$\beta = \mathbb{E}_{\mathbf{x}, \epsilon}[\mathbf{y}] = \int \mathbf{y} d\mu_{\mathbf{x}}(x) d\mu_{\epsilon}(\epsilon).$$

Each component function $\mathbf{f}^{(k)}$ is mean-zero with respect to the marginal of $\mathbf{x}^{(k)}$.

Under such signal structure, we label each machine at boosting round b using feature k in Algorithm 1 as $\hat{\mathbf{f}}^{(k)}(\mathbf{x})$.

3 Assumption

1. **Additive Lipschitz Nonparametric Regression:** Let the response satisfy

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where $\mathbf{y}, \beta \in \mathbb{R}$, $\mathbf{x} = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(p)})^\top \in \mathbb{R}^p$, and ϵ are i.i.d. sub-Gaussian errors with mean zero and variance $\sigma^2 < \infty$. Assume the covariates have a bounded density on their support: $\exists 0 < c_1 \leq \mu(\mathbf{x}) \leq c_2 < \infty$. Each component function $\mathbf{f}^{(k)} : \mathbb{R} \rightarrow \mathbb{R}$ is mean-zero with respect to the marginal of $\mathbf{x}^{(k)}$, and is L -Lipschitz continuous. The intercept vector β is defined as the unconditional mean of the response,

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Each component function $\mathbf{f}^{(k)}$ is mean-zero with respect to the marginal of $\mathbf{x}^{(k)}$.

2. **Structure-Value Isolation:** The structure of any tree is independent of the prediction made by the terminal nodes. Algebraically, for a single tree at iteration b for feature k , one can write the prediction on a training set $\mathbf{X}$ as a linear map from

the observed vector $\mathbf{y}$ to the prediction: $\hat{\mathbf{t}}(\mathbf{x}) = \mathbf{S}_b^{(k)} \mathbf{y}$, where $\mathbf{S}_b^{(k)}$ is the tree structure matrix defined below:

$$\mathbf{S}_{i,j}^{(k)} = \left[\frac{\mathbf{I}(\mathbf{x}_j^{(k)} \in A_i) \mathbf{I}(j \in w)}{\sum_{l=1}^n \mathbf{I}(\mathbf{x}_l^{(k)} \in A_i) \mathbf{I}(l \in w)} \right]$$

3. **Non-adaptivity** There exists a measure $\mathcal{Q}_n^{(k)}$ on the space of tree structures such that after some burn-in period b' , each individual univariate tree structure constructed using feature k is an independent draw from $\mathcal{Q}_n^{(k)}$.
4. **Partition Distinctness:** After the burn-in period, let $\mathbb{E}[\mathbf{S}^{(k)}]$ denote the expected structure matrix for trees trained on feature k under $\mathcal{Q}_n^{(k)}$. We assume there exists a constant $c_3 < 1$ such that for all distinct features $a \neq k$, $c_3(p-1) < 1$ and

$$\|\mathbb{E}[\mathbf{S}^{(a)}] \mathbb{E}[\mathbf{S}^{(k)}]\| \leq c_3.$$

5. **Shrinking Volume** For any $\nu > 0$, the minimal leaf size is shrinking at rate $v_n = n^{-\frac{d+1}{d+2}+\nu} < n^{-\frac{d}{d+1}} := O(d_n^d)$.
6. **Asymptotic Locality** We introduce a sequence $\{d_n\}$ with $d_n > 0$, $d_n \rightarrow 0$, by which the diameter of any leaf is bounded from above. We require

$$\sup_{A \in \mathcal{Q}_n} \text{diam}(A) = O(d_n), \quad d_n = O\left(n^{-\frac{1}{d+1}}\right).$$

7. **Increasing Leaf** The minimal number of points within one leaf is lower bounded by

$$\inf_{A \in \Pi \in \mathcal{Q}_n^{(k)}} \sum_{i=1}^n \mathbf{I}(\mathbf{x}_{i,n} \in A) = \Omega(n^{\frac{1}{d+2}})$$

4 Algorithms

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Algorithm 1 Inferable EBM

- 1: **Input:** $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\mathbf{y} \in \mathbb{R}^n$, subsample rate ξ , rounds B , truncation level $M > 0$.
 - 2: **Init:** $\beta_0 \leftarrow \frac{1}{n} \sum_{i=1}^n y_i$; $f_k^{(0)} \leftarrow 0$ for all k .
 - 3: **for** $b = 1$ **to** B **do**
 - 4: **for** $k = 1$ **to** p **in parallel do**
 - 5: Sample $G_{b,k} \subset \{1, \dots, n\}$ i.i.d. each i w.p. ξ .
 - 6: $r_{i,k} \leftarrow y_i - \left(\beta_0 + \sum_{\ell} f_{\ell}^{(b-1)}(x_i^{(\ell)}) \right)$ for $i \in G_{b,k}$.
 - 7: Fit tree $t^{(b,k)}$ to $\{(x_i^{(k)}, r_{i,k})\}_{i \in G_{b,k}}$.
 - 8: $\mu_{b,k} \leftarrow \frac{1}{n} \sum_{i=1}^n t^{(b,k)}(x_i^{(k)})$.
 - 9: $\tilde{t}^{(b,k)}(x) \leftarrow t^{(b,k)}(x) - \mu_{b,k}$.
 - 10: $\tilde{t}^{(b,k)}(x) \leftarrow \max\{-M, \min\{\tilde{t}^{(b,k)}(x), M\}\}$ for all x .
 - 11: $\beta_0 \leftarrow \beta_0 + \mu_{b,k}$.
 - 12: $f_k^{(b)} \leftarrow \frac{b-1}{b} f_k^{(b-1)} + \frac{1}{b} \tilde{t}^{(b,k)}$.
 - 13: **end for**
 - 14: **end for**
 - 15: **Output:** $\hat{f}(x) = \beta_0 + \sum_{k=1}^p f_k^{(B)}(x^{(k)})$ with $\sum_i f_k^{(B)}(x_i^{(k)}) = 0$ for all k .
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A Finite Sample Convergence

A.1 A Fixed-point Iteration

To investigate the finite sample convergence behavior of the algorithm, we begin with a set of postulated fixed point. For each subensemble's prediction on the training set, assume the existence of a fixed point $\tilde{\mathbf{y}}_k^*$. Then the BREBM update rule leads to the following fixed point iteration:

$$\tilde{\mathbf{y}}_k^* = \frac{1}{b+1} \left(\left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} (Y - \sum_{a \neq k} \tilde{\mathbf{y}}_a^*) + b \tilde{\mathbf{y}}_k^* \right)$$

Rearranging the equation gives the following identity

$$\tilde{\mathbf{y}}_k^* = (I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} (\mathbf{y} - \hat{\mathbf{y}}) \quad (1)$$

Since we do not know $\hat{\mathbf{y}}$, we will solve for it and plug it back later. Sum the equations over k yields:

$$\hat{\mathbf{y}} := \sum_{k=1}^p \tilde{\mathbf{y}}_k^* = \sum_{k=1}^p (I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} (\mathbf{y} - \hat{\mathbf{y}})$$

which represents our ensemble prediction as a kernel ridge regression, with the group-level kernel being a sum of feature-wise kernel ridge operator (i.e. $S_n = \sum_{k=1}^p (I - S_n^{(k)})^\dagger S_n^{(k)}$)

$$\hat{\mathbf{y}} = \left(I + \sum_{k=1}^p (I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} \right)^\dagger \left(\sum_{k=1}^p (I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} \right) \mathbf{y}$$

Plugging it back to equation 1 produces the fixed-points for each subensemble:

$$\begin{aligned} \tilde{\mathbf{y}}_k^* &= (I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} \left[I - \left(I + \sum_{k=1}^p (I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} \right)^\dagger \left(\sum_{k=1}^p (I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} \right) \right] \mathbf{y} \\ &:= (I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)} R \mathbf{y} \end{aligned}$$

The operator $(I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)}$, which we can interpret as a feature-wise KRR, is stacked with the fixed point of the residual created on the whole ensemble. In this product, $(I - S_n^{(k)})^\dagger \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) S_n^{(k)}$ measures the marginal contribution of a particular feature k contribute to the ensemble prediction.

A.2 Specialized Tree Kernels

Backfitting is known to fail under heavy multicollinearity: the ensemble predictions may converge, but the feature-wise functionals do not, due to non-identifiability. In our setting, this manifests as large overlaps between the kernel spaces of different features. To guarantee convergence, we must ensure that the residuals from other features are not fully contained in the kernel space of the current feature. When features are independent, this holds automatically; when they are correlated, it suffices to bound the cross-feature overlap strictly below one.

In practice, we enforce the centering condition

$$\sum_{i=1}^n f_k(x_i^{(k)}) = 0, \quad \forall k,$$

which removes the global constant direction, thereby eliminating the trivial overlap between feature-specific structure matrices. Formally, we state the setup below.

Consider two features $X^{(a)}$ and $X^{(k)}$. Define the maximal correlation

$$\rho_*(X^{(a)}, X^{(k)}) = \sup_{\substack{g, h \\ \mathbb{E}[g] = \mathbb{E}[h] = 0 \\ \mathbb{E}[g^2] = \mathbb{E}[h^2] = 1}} \mathbb{E}[g(X^{(a)}) h(X^{(k)})].$$

This quantity equals the operator norm of the conditional-expectation projections: $\|P_a P_k\| = \rho_*(X^{(a)}, X^{(k)})$, where $P_k f = \mathbb{E}[f \mid X^{(k)}]$ is the L^2 projection onto functions of feature k . In particular, if features are not perfectly collinear, then $\rho_*(X^{(a)}, X^{(k)}) < 1$.

In the finite-sample case, a tree grown on feature k induces a partition Π_k of the sample indices. Its structure matrix S_{Π_k} is the orthogonal projector in $\mathbb{R}^n$ onto vectors constant on each leaf. Under SVI and Non-adaptivity, averaging over trees yields an expected structure matrix $\mathbb{E}[S^{(k)}]$ that depends only on the marginal distribution of $X^{(k)}$. As $n \rightarrow \infty$, these expected matrices converge to the population projection P_k . Hence after burn-in, the overlap between two expected structure matrices is asymptotically controlled by the maximal correlation:

$$\|\mathbb{E}[S^{(a)}] \mathbb{E}[S^{(k)}]\| \leq \rho_*(X^{(a)}, X^{(k)}) + o_n(1).$$

Thus, no perfect collinearity implies a strict bound < 1 . A sufficient margin for contraction is

$$\sup_{a \neq k} \rho_*(X^{(a)}, X^{(k)}) < \frac{1}{p-1},$$

which ensures $\|\mathbf{L}\| \leq (p-1)c_3 < 1$ in the block-operator bound. Geometrically, we are looking at two partitions that does not vote more than $\frac{n}{p-1}$ points in same leaves, which suggest a uniform expansion of our observed data points in regions as we move into higher dimensions. Equivalently, one may assume the row-sum incoherence condition

$$\max_k \sum_{a \neq k} \|\mathbb{E}[S^{(k)}] \mathbb{E}[S^{(a)}]\| < 1,$$

A.3 Contraction

Under stated assumptions, we can prove that Algorithm 1's prediction converge almost surely to a postulated fixed point presented in Equation ???. Formally, we make use of Theorem C.1, called a stochastic contraction mapping theorem, to demonstrate the difference between the current prediction and the fixed point is a stochastic contraction mapping.

Theorem A.1. *Denote the prediction using feature k at boosting round b as $\hat{\mathbf{y}}_b^{(k)} = \frac{1}{b} \sum_{s=1}^b t^{(s,k)}(\mathbf{x}^{(k)})$. Define the aggregated kernel $\mathcal{K}_n := \sum_{k=1}^p (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^\dagger (\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top) \mathbb{E}\mathbf{S}^{(k)}$. Suppose for each feature, there exists a fixed point:*

$$\tilde{\mathbf{y}}_k^* = (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^\dagger (\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top) \mathbb{E}\mathbf{S}^{(k)} \left[\mathbf{I} - \left(\mathbf{I} + \mathcal{K}_n \right)^\dagger \mathcal{K}_n \right] \mathbf{y}$$

which implies a global fixed point

$$\hat{\mathbf{y}}^* = \hat{\beta}^* + \mathcal{K}_n \left[\mathbf{I} - \left(\mathbf{I} + \mathcal{K}_n \right)^\dagger \mathcal{K}_n \right] \mathbf{y}$$

Then $\hat{\mathbf{y}}_b^{(k)} \xrightarrow{a.s.} \tilde{\mathbf{y}}_k^*$ and $\hat{\mathbf{y}}_b \xrightarrow{a.s.} \hat{\mathbf{y}}^*$ under stated assumptions in 3. In the meantime, $\hat{\beta}^* \xrightarrow{a.s.} \bar{\mathbf{y}} := \frac{1}{n} \mathbf{1}^\top \mathbf{y}$.

Proof. Denote $\mathbf{D}_b^{(k)} = \hat{\mathbf{y}}_b^{(k)} - \tilde{\mathbf{y}}_k^*$. Define the stacked distance vector $\mathbf{D}_b = [\mathbf{D}_b^{(1)}, \dots, \mathbf{D}_b^{(p)}]^\top \in \mathbb{R}^{np}$. Showing this random vector is a stochastic contraction mapping suffices to establish coordinate-wise convergence.

For each coordinate, the conditional mean satisfies

$$\begin{aligned}
 \mathbb{E}[\mathbf{D}_{b+1}^{(k)} \mid \mathcal{F}_b] &= \mathbb{E}[\hat{\mathbf{y}}_{b+1}^{(k)} - \tilde{\mathbf{y}}_k^* \mid \mathcal{F}_b] \\
 &= \mathbb{E}\left[\frac{b}{b+1}\hat{\mathbf{y}}_b^{(k)} + \frac{1}{b+1}\left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbf{S}_n^{b+1,(k)}(\mathbf{y} - \Gamma_M\left(\sum_{a \neq k}\hat{\mathbf{y}}_b^{(a)}\right)) - \tilde{\mathbf{y}}_k^* \mid \mathcal{F}_b\right] \\
 &= \frac{b}{b+1}\hat{\mathbf{y}}_b^{(k)} + \frac{1}{b+1}\left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbb{E}[\mathbf{S}_n^{b+1,(k)} \mid \mathcal{F}_b](\mathbf{y} - \Gamma_M\left(\sum_{a \neq k}\hat{\mathbf{y}}_b^{(a)}\right)) - \tilde{\mathbf{y}}_k^* \\
 &= \frac{b}{b+1}\mathbf{D}_b^{(k)} + \frac{1}{b+1}\left(\left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbb{E}[\mathbf{S}_n^{b+1,(k)} \mid \mathcal{F}_b](\mathbf{y} - \Gamma_M\left(\sum_{a \neq k}\hat{\mathbf{y}}_b^{(a)}\right)) - \tilde{\mathbf{y}}_k^*\right) \\
 &= \frac{b}{b+1}\mathbf{D}_b^{(k)} + \frac{1}{b+1}\left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)(\mathbb{E}[\mathbf{S}_n^{(b,k)} \mid \mathcal{F}_b] - \mathbb{E}[\mathbf{S}_n^{(\infty,k)} \mid \mathcal{F}_b])(\mathbf{y} - \sum_{a \neq k}\tilde{\mathbf{y}}_a^*) \\
 &\quad + \frac{1}{b+1}\left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbb{E}[\mathbf{S}_n^{(b,k)} \mid \mathcal{F}_b](\sum_{a \neq k}\tilde{\mathbf{y}}_a^* - \Gamma_M\left(\sum_{a \neq k}\hat{\mathbf{y}}_b^{(a)}\right)) \\
 &= \frac{b}{b+1}\mathbf{D}_b^{(k)} + \frac{1}{b+1}\left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbb{E}[\mathbf{S}_n^{(b,k)} \mid \mathcal{F}_b](\sum_{a \neq k}\tilde{\mathbf{y}}_a^* - \Gamma_M\left(\sum_{a \neq k}\hat{\mathbf{y}}_b^{(a)}\right)) \\
 &:= \frac{b}{b+1}\mathbf{D}_b^{(k)} + \frac{1}{b+1}\left(I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbb{E}[\mathbf{S}_n^{(b,k)} \mid \mathcal{F}_b]\Delta_b^{(-k)}
 \end{aligned}$$

where $\|\Delta_b^{(-k)}\| \leq \|\sum_{a \neq k}\mathbf{D}_b^{(a)}\|$. Without truncation, stacking across all features yields

$$\mathbb{E}[\mathbf{D}_{b+1} \mid \mathcal{F}_b] = \left(\frac{b}{b+1}\mathbf{I}_{np} + \frac{1}{b+1}\mathbf{L}\right)\mathbf{D}_b,$$

where $\mathbf{L}$ is the $np \times np$ block matrix with diagonal blocks equal to 0 and off-diagonal blocks given by $\mathbb{E}\mathbf{S}^{(k)}$, multiplied by the averaging operator $(I - \frac{1}{n^2}\mathbf{1}\mathbf{1}^\top)$.

The Partition Distinctness assumption implies that each off-diagonal block of $\mathbf{L}$ has operator norm at most c_3 , so by a Schur-type bound

$$\|\mathbf{L}\| \leq (p-1)c_3 < 1$$

Consequently the stacked update satisfies

$$\|\mathbb{E}[\mathbf{D}_{b+1} \mid \mathcal{F}_b]\| \leq \frac{b + \|\mathbf{L}\|}{b+1}\|\mathbf{D}_b\|$$

For the deviation condition, if leaf predictions are bounded by M , then $\|\epsilon_{b+1}\| \leq \frac{2M\sqrt{p}}{b+1}$, so $\sup_b \|\epsilon_b\| \rightarrow 0$ and $\sum_b \mathbb{E}[\|\epsilon_b\|^2] < \infty$. This verifies the stochastic contraction mapping conditions.

It remains to verify the convergence of the intercept term $\hat{\beta}$. The update rule for the intercept is given by

$$\hat{\beta}_{b+1} := \hat{\beta}_b + \frac{1}{n} \sum_{k=1}^p \sum_{i=1}^n \hat{\mathbf{f}}_b^{(k)}(x_i)$$

To identify this limit, recall that at the postulated fixed point we have

$$\hat{\mathbf{y}}^* = \hat{\beta}^* + \sum_{k=1}^p \tilde{\mathbf{y}}_k^*,$$

with $\frac{1}{n}\mathbf{1}^\top \tilde{\mathbf{y}}_k^* = 0$ for all k . Taking the average over i gives

$$\frac{1}{n}\mathbf{1}^\top \hat{\mathbf{y}}^* = \hat{\beta}^* + \frac{1}{n} \sum_{k=1}^p \mathbf{1}^\top \tilde{\mathbf{y}}_k^* = \hat{\beta}^*.$$

But $\frac{1}{n}\mathbf{1}^\top \hat{\mathbf{y}}^* = \frac{1}{n}\mathbf{1}^\top \mathbf{y} = \bar{y}$ since the residuals at convergence have mean zero. Therefore the intercept converges almost surely to the global mean of the response,

$$\hat{\beta}_b \xrightarrow{a.s.} \bar{y} = \frac{1}{n}\mathbf{1}^\top \mathbf{y}.$$

□

We drop the subscript n if no ambiguity is caused.

Corollary A.1. *By defining $\hat{\mathbf{f}}^{(k)} := \lim_{b \rightarrow \infty} \hat{\mathbf{f}}_b^{(k)}$, the feature-wise prediction:*

$$\hat{\mathbf{f}}^{(k)}(\mathbf{x}) = \mathbb{E}[\mathbf{s}^{(k)}(\mathbf{x})]^\top (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^\dagger (\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top) \left[\mathbf{I} - (\mathbf{I} + \mathcal{K}_n)^\dagger \mathcal{K}_n \right] \mathbf{y}$$

and the ensemble prediction

$$\hat{\mathbf{f}}(\mathbf{x}) := \bar{\mathbf{y}} + \sum_{k=1}^p \mathbb{E}[\mathbf{s}^{(k)}(\mathbf{x})]^\top (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^\dagger (\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top) \left[\mathbf{I} - (\mathbf{I} + \mathcal{K}_n)^\dagger \mathcal{K}_n \right] \mathbf{y}$$

Intuitively, after centering each signal vector, we would have a decomposition with additive feature-shaped functionals plus a intercept, which is $\bar{\mathbf{y}}$. If we plot the partial-dependency figure, we would see all predicted signal oscillating around 0. In that case, interpretation is as follow: If we observe no features, the best guess would be the baseline $\hat{\beta}^* = \bar{y}$. As we adding more and more features, we are able to recover greater parts of the function landscape.

B Limiting Distribution

In Section A, we have shown that as we keep boosting, our feature estimators will converge to a stacked kernel-ridge-regression

$$\hat{\mathbf{y}}^* = \hat{\beta}^* + \mathcal{K} \left[\mathbf{I} - (\mathbf{I} + \mathcal{K})^\dagger \mathcal{K} \right] \mathbf{y}$$

with

$$\mathcal{K} := \sum_{k=1}^p (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^\dagger (\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top) \mathbb{E}\mathbf{S}^{(k)}$$

Suppose the true underlying function is a general additive model of the form:

$$\mathbf{f}(\mathbf{x}) = \beta + \sum_{k=1}^p \mathbf{f}^{(k)}(\mathbf{x})$$

where $\beta = \mathbb{E}_{\mathbf{x}}[\mathbf{f}(\mathbf{x})]$, supporting the signal as a baseline, with other functionals centered around 0. Starting here, we will use abbreviations below if no ambiguity:

$$\mathbf{k}^{(k)}(\mathbf{x}) := \mathbb{E}[\mathbf{s}^{(k)}(\mathbf{x})], \quad \mathbf{K}^{(k)} := \mathbb{E}[\mathbf{S}^{(k)}], \quad \mathbf{r}^{(k)}(\mathbf{x}) := \mathbf{k}^{(k)}(\mathbf{x})^\top (\mathbf{I} - \mathbf{K}^{(k)})^\dagger (\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top) \left[\mathbf{I} - (\mathbf{I} + \mathcal{K})^\dagger \mathcal{K} \right]$$

So our out-of-sample prediction given any $\mathbf{x}$ is $\hat{\mathbf{f}}^{(k)}(\mathbf{x}) = \mathbf{r}^{(k)}(\mathbf{x})^\top \mathbf{y}$.

To show asymptotic normality, consider the following decomposition:

$$\frac{\widehat{\mathbf{f}}_n^{(k)}(\mathbf{x}) - \mathbf{f}^{(k)}(\mathbf{x})}{\|\mathbf{r}^{(k)}\|} = \frac{\widehat{\mathbf{f}}_n^{(k)}(\mathbf{x}) - \langle \mathbf{r}^{(k)}(\mathbf{x}), \mathbf{f}(\mathbf{X}_n) \rangle}{\|\mathbf{r}^{(k)}\|} + \frac{\langle \mathbf{r}^{(k)}(\mathbf{x}), \mathbf{f}(\mathbf{X}_n) \rangle - \mathbf{f}^{(k)}(\mathbf{x})}{\|\mathbf{r}^{(k)}\|} \quad (2)$$

The first term can be easily shown by constraining leaf size, which we discuss in Section B.1. The later term is trickier. Generalizing from training set to the whole input space, we expect a convergence that is at least of a weak. Bounded kernel, fortunately, allows us to do so.

B.1 Conditional Asymptotic Normality

We begin analysis by showing asymptotic normality on the training set. Formally, denote $\mathbf{f}(\mathbf{X}) = [\mathbf{f}(\mathbf{x}_1), \dots, \mathbf{f}(\mathbf{x}_n)]^\top$. What we want to show is that, given any $\mathbf{x} \in \mathbb{R}^p, \forall k$:

$$\frac{\widehat{\mathbf{f}}^{(k)}(\mathbf{x}) - \mathbf{r}^{(k)}(\mathbf{x})^\top \mathbf{f}(\mathbf{X})}{\|\mathbf{r}^{(k)}(\mathbf{x})\|} \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

The key component of this proof is the rate of $\|\mathbf{r}^{(k)}\|_1, \|\mathbf{r}^{(k)}\|, \|\mathbf{r}^{(k)}\|_\infty$. These rates will be built on a result shown in the original Boulevard paper on the behavior of subsampling rate.

Lemma B.1. *Considering a subsampled regression tree. Assume that each leaf contains no fewer than $n^{\frac{1}{p+2}}$ sample points before subsampling. If we are subsampling at rate at least $\xi = n^{-\frac{1}{p+2} \log n}$, then the expected structure vector's norm follows the rate below:*

$$\|\mathbf{k}^{(k)}\| = 1 - O\left(\frac{1}{n}\right)$$

Our plan to conditional CLT is as follow: Lemma B.1 specifies a proper subsample rates that enables norm convergence. Lindeberg condition simply asks us to check the truncation level $\frac{\mathbf{r}_i^{(k)}}{\|\mathbf{r}^{(k)}\|}$. We will first study the norm of $\mathbf{k}^{(k)}$. To bound the ℓ_1 norm, it suffices to imposing conditions on the leaf size (with respect to the Lebesgue measure of data points). To bound the ℓ_2 norm. A simple algebra utilizing the rate of ℓ_1 norm will help with that, too. If this norm is well controlled, notice that the Boulevard vector $\mathbf{r}^{(k)}$ is simply a sum of rows in the matrix $\mathbf{P} := (\mathbf{I} - \mathbf{K}^{(k)})^\dagger (\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top) \left[\mathbf{I} - (\mathbf{I} + \mathcal{K})^\dagger \mathcal{K} \right]$ with the weighting vector being $\mathbf{k}^{(k)}$. Hence it suffices to study the eigenvalues of the operator $\mathbf{P}$, which we discuss below.

Lemma B.2 (Spectrum of $\mathbf{K}^{(k)}$). *For any feature k , the expected structure matrix $\mathbf{K}^{(k)} = \mathbb{E}[\mathbf{S}^{(k)}]$ is a symmetric operator on $\mathbb{R}^n$ with spectrum contained in $[0, 1]$. It always holds that $\mathbf{K}^{(k)}\mathbf{1} = \mathbf{1}$, so 1 is an eigenvalue with eigenvector $\mathbf{1}$. Restricting to the mean-zero subspace $\mathbf{H}_0 = \{v : \mathbf{1}^\top v = 0\}$, the supremum of the spectrum is bounded by*

$$\sup\{\lambda : \lambda \in \sigma(\mathbf{K}^{(k)}|_{\mathbf{H}_0})\} = \rho_k < 1,$$

provided the distribution of partitions under $\mathcal{Q}_n^k$ is not almost surely degenerate.

Proof. Fix any unit $v \in \mathbf{H}_0$. Because $\mathbf{S}^{(k)}$ is an orthogonal projector, $v^\top \mathbf{S}^{(k)} v = \|\mathbf{S}^{(k)} v\|^2$, with equality $\|\mathbf{S}^{(k)} v\| = 1$ if and only if $v \in \text{range}(\mathbf{S}^{(k)})$. Hence

$$v^\top \mathbf{K}^{(k)} v = \mathbb{E}[v^\top \mathbf{S}^{(k)} v] = \mathbb{E}[\|\mathbf{S}^{(k)} v\|^2] \leq 1.$$

Assume toward a contradiction that $\lambda_{\max}(\mathbf{K}^{(k)}|_{\mathbf{H}_0}) = 1$. Then there exists a sequence of unit vectors $v_m \in \mathbf{H}_0$ with $v_m^\top \mathbf{K}^{(k)} v_m \rightarrow 1$. By the display, $\mathbb{E}[\|\mathbf{S}^{(k)} v_m\|^2] \rightarrow 1$. The map $\Pi \mapsto \|\mathbf{S}^{(k)} \Pi\|^2$ takes values in $[0, 1]$, so if its expectation tends to 1, we must have $\|\mathbf{S}^{(k)} v_m\|^2 \rightarrow 1$ in probability (with respect to m). Passing to a subsequence if necessary and using compactness of the unit sphere, take $v_m \rightarrow v_\star$ with $\|v_\star\| = 1$ and $v_\star \in \mathbf{H}_0$. By lower semicontinuity and the projector property, $\|\mathbf{S}^{(k)} v_\star\| = 1$ almost surely, hence $\mathbf{S}^{(k)} v_\star = v_\star$ almost surely. Therefore $v_\star \in \text{range}(\mathbf{S}^{(k)})$ for almost every draw of $\mathbf{S}^{(k)}$.

If the distribution of $\mathbf{S}^{(k)}$ charges at least two distinct partitions with positive probability, their ranges (as subspaces of $\mathbb{R}^n$) intersect on $\mathbf{H}_0$ only in $\{0\}$ unless $v_\star$ is proportional to $\mathbf{1}$, which it is not. Thus $v_\star$ cannot lie in the range of both with positive probability, a contradiction. Hence $\lambda_{\max}(\mathbf{K}^{(k)}|_{\mathbf{H}_0}) < 1$. $\square$

Lemma B.2 gives a strict bound on the eigenvalues of the expected kernel, which we can utilize to stabilize the behavior of the aggregated kernel $\mathcal{K}$, which is described in the following lemma.

Lemma B.3 (Aggregated kernel). *Work on the mean-zero subspace $\mathbf{H}_0 = \{v \in \mathbb{R}^n : \mathbf{1}^\top v = 0\}$. The aggregated kernel*

$$\mathcal{K} := \sum_{k=1}^p (\mathbf{I} - \mathbf{K}^{(k)})^\dagger \left(\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right) \mathbf{K}^{(k)}$$

is symmetric and positive semidefinite. Since $(\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top)$ is the identity on $\mathbf{H}_0$, we may write on $\mathbf{H}_0$ simply $\mathcal{K} = \sum_k (\mathbf{I} - \mathbf{K}^{(k)})^\dagger \mathbf{K}^{(k)}$. If $\rho_k := \|\mathbf{K}^{(k)}\|_{\mathbf{H}_0} < 1$, then on $\mathbf{H}_0$ one has the operator sandwich

$$\sum_{k=1}^p \mathbf{K}^{(k)} \preceq \mathcal{K} \preceq \sum_{k=1}^p \frac{1}{1 - \rho_k} \mathbf{K}^{(k)},$$

hence $\lambda_{\min}^+(\mathcal{K}) \geq \lambda_{\min}^+(\sum_a \mathbf{K}^{(a)})$ and $\lambda_{\max}(\mathcal{K}) \leq \sum_k \frac{\rho_k}{1 - \rho_k}$.

Proof. Each summand $(\mathbf{I} - \mathbf{K}^{(k)})^\dagger \mathbf{K}^{(k)}$ is symmetric because $\mathbf{K}^{(k)}$ is symmetric and $(\mathbf{I} - \mathbf{K}^{(k)})^\dagger$ is a Borel function of $\mathbf{K}^{(k)}$; positivity follows from $x \mapsto (1 - x)^\dagger x$ being nonnegative on $[0, 1]$. On $\mathbf{H}_0$, the eigenvalues of $(\mathbf{I} - \mathbf{K}^{(k)})^\dagger$ lie in $[1, 1/(1 - \rho_k)]$, so for any $v \in \mathbf{H}_0$,

$$v^\top \mathbf{K}^{(k)} v \leq v^\top (\mathbf{I} - \mathbf{K}^{(k)})^\dagger \mathbf{K}^{(k)} v \leq \frac{1}{1 - \rho_k} v^\top \mathbf{K}^{(k)} v.$$

Summing over k gives the stated Loewner bounds, from which the eigenvalue bounds follow. $\square$

Combining those bounds above, we are able to claim that our EBM operator $\mathbf{P}$ will not explode in some directions. Formally, we have the following lemma.

Lemma B.4 (Eigenvalues of $\mathbf{P}$). *Work on the mean-zero subspace $\mathbf{H}_0 = \{v \in \mathbb{R}^n : \mathbf{1}^\top v = 0\}$. Define*

$$\mathbf{P} := (\mathbf{I} - \mathbf{K}^{(k)})^\dagger \left(\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right) \left[\mathbf{I} - (\mathbf{I} + \mathcal{K})^\dagger \mathcal{K} \right], \quad \mathcal{K} := \sum_{k=1}^p (\mathbf{I} - \mathbf{K}^{(k)})^\dagger \left(\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right) \mathbf{K}^{(k)}.$$

On $\mathbf{H}_0$ the centering projector is the identity and $(\mathbf{I} + \mathcal{K})^\dagger = (\mathbf{I} + \mathcal{K})^{-1}$, hence

$$\mathbf{P} = (\mathbf{I} - \mathbf{K}^{(k)})^\dagger (\mathbf{I} + \mathcal{K})^{-1} \quad \text{on } \mathbf{H}_0.$$

The matrix $\mathbf{P}$ is symmetric and positive semidefinite on $\mathbf{H}_0$. Writing $\rho_k := \|\mathbf{K}^{(k)}\|_{\mathbf{H}_0} < 1$, its eigenvalues lie in the interval

$$\frac{1}{1 + \lambda_{\max}(\mathcal{K})} \leq \lambda(\mathbf{P}) \leq \frac{1}{(1 - \rho_k)(1 + \lambda_{\min}^+(\mathcal{K}))}.$$

In particular, $\|\mathbf{P}\| \leq ((1 - \rho_k)(1 + \lambda_{\min}^+(\mathcal{K})))^{-1} \leq (1 - \rho_k)^{-1}$. If, in addition, $\mathbf{K}^{(k)}$ and $\mathcal{K}$ commute on $\mathbf{H}_0$, then $\mathbf{P}$ is diagonalizable in a common eigenbasis and

$$\lambda_i(\mathbf{P}) = \frac{1}{1 - \lambda_i(\mathbf{K}^{(k)})} \cdot \frac{1}{1 + \lambda_i(\mathcal{K})}, \quad i = 1, \dots, n - 1.$$

Proof. On $\mathbf{H}_0$ we have $\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top = \mathbf{I}$ and

$$\mathbf{I} - (\mathbf{I} + \mathcal{K})^\dagger \mathcal{K} = \mathbf{I} - (\mathbf{I} + \mathcal{K})^{-1} \mathcal{K} = (\mathbf{I} + \mathcal{K})^{-1},$$

so $\mathbf{P} = (\mathbf{I} - \mathbf{K}^{(k)})^\dagger (\mathbf{I} + \mathcal{K})^{-1}$ on $\mathbf{H}_0$. Since $\mathbf{K}^{(k)}$ and $\mathcal{K}$ are symmetric, by the spectral theorem we may write $\mathbf{K}^{(k)} = U \Lambda U^\top$ with $\Lambda = \text{diag}(\lambda_i)$ and define $(\mathbf{I} - \mathbf{K}^{(k)})^\dagger = U(\mathbf{I} - \Lambda)^\dagger U^\top$, where $(1 - \lambda_i)^\dagger = 1/(1 - \lambda_i)$ if $\lambda_i \neq 1$ and 0 otherwise. Thus $(\mathbf{I} - \mathbf{K}^{(k)})^\dagger$ is symmetric and positive semidefinite on $\mathbf{H}_0$, with eigenvalues in $[1, (1 - \rho_k)^{-1}]$. Likewise, $\mathcal{K}$ is symmetric positive semidefinite, so $(\mathbf{I} + \mathcal{K})^{-1}$ is symmetric positive definite on $\mathbf{H}_0$, with eigenvalues in $[1/(1 + \lambda_{\max}(\mathcal{K})), 1/(1 + \lambda_{\min}^+(\mathcal{K}))]$. The product of symmetric positive semidefinite matrices has nonnegative spectrum; moreover, for any matrices A, B the extremal singular values satisfy $s_{\min}(AB) \geq s_{\min}(A)s_{\min}(B)$ and $s_{\max}(AB) \leq s_{\max}(A)s_{\max}(B)$. Applying these to $A = (\mathbf{I} - \mathbf{K}^{(k)})^\dagger$ and $B = (\mathbf{I} + \mathcal{K})^{-1}$ yields the stated eigenvalue interval and the norm bound. If $\mathbf{K}^{(k)}$ and $\mathcal{K}$ commute, they are simultaneously diagonalizable on $\mathbf{H}_0$, and the eigenvalue formula follows by multiplying the diagonal entries. $\square$

Then our analysis proceed with the assistance of the following theorem, cited from the original Boulevard paper as well.

Lemma B.5 (Rate of Convergence). *Let $\mathbf{B}_n := \{i : \|\mathbf{x} - \mathbf{x}_i\| \leq d_n\}$ be the points within distance d_n from test point $\mathbf{x}$. If $|\mathbf{B}_n| = \Omega(n \cdot d_n^d)$, then $\|\mathbf{k}^{(k)}\|_2 = \Theta(n^{-\frac{1}{2} \frac{1}{d+1}})$, $\|\mathbf{r}^{(k)}\|_2 = \Theta(n^{-\frac{1}{2} \frac{1}{d+1}})$.*

This lemma holds true regardless of the design of our boosting process. The key argument is the minimal leaf size. A sketch of proof is appended below.

Proof. To bound $\|\mathbf{k}^{(k)}\|$, recall:

$$\mathbf{k}_{nj}^{(k)} = \mathbb{E}[\mathbf{s}_{n,j}^{(k)}(x)] = \mathbb{E} \left[\frac{\mathbf{1}(x_j \in A)}{\sum_{j=1}^n \mathbf{1}(x_j \in A)} \right], x \in A$$

encodes the expected influence of point x_j on a point of interest x among other n points. Then the condition

$$\inf_{A \in \mathcal{Q}_n} \sum_{i=1}^n \mathbf{1}(x_i \in A) \geq \Omega(n^{\frac{1}{d+1}})$$

implies that $\mathbf{k}_{nj}^{(k)} = \mathcal{O}n^{-\frac{1}{d+1}}$, since we can't have total weights of more than $\mathcal{O}(1)$ in a leaf. By Lemma B.1, $\|\mathbf{k}^{(k)}\|_1 \leq 1$,

$$\|\mathbf{k}^{(k)}\| \leq \sqrt{\|\mathbf{k}^{(k)}\|_1 \|\mathbf{k}^{(k)}\|_\infty} = \mathcal{O}n^{-\frac{1}{2} \frac{1}{d+1}}.$$

By assestion $|\mathbf{B}_n| = \Omega(n \cdot d_n^d)$, there are at most

$$\Omega(n \cdot d_n^d) = \Omega(n^{\frac{1}{d+1}})$$

k_{nj} 's that are positive. Equivalently, we can have the magnitude of each non-zero $\mathbf{k}_{nj}^{(k)}$ is lower bounded by $\Omega(n^{-\frac{1}{d+1}})$. This holds also because the total weights can't be larger than $\mathcal{O}(1)$. Since $\|\mathbf{k}^{(k)}\|_1 = 1 - \mathcal{O}(n^{-1})$,

$$\|\mathbf{k}^{(k)}\| = \Omega(\sqrt{(n^{-\frac{1}{d+1}})^2 \cdot n^{\frac{1}{d+1}}}) = \Omega(n^{-\frac{1}{2} \frac{1}{d+1}}).$$

By lemma B.4, we know that $\mathbf{r}^{(k)}$ is a linear transformation from $\mathbf{k}^{(k)}$ by $\mathbf{P}$, where $\mathbf{P}$ has bounded eigenvalues. Hence $\|\mathbf{r}^{(k)}\|$ enjoys similar rates. $\square$

Having access to the rate of convergence of $\mathbf{r}^{(k)}$, we can fill in the blank of the weighting vector argument in the Lindeberg conditions. Concretely, it controls any error deviate too much from 0 that contributes to non-normality. We state the theorem below.

Theorem B.1 (Conditional Asymptotic Normality). *For any $\mathbf{x} \in [0, 1]^p$, write $f(\mathbf{X}_n) = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))^\top$. Then under SVI, Non-Adaptivity, Minimal Leaf Size and assumptions in Lemma B.5, we have*

$$\frac{\widehat{\mathbf{f}}_n^{(k)}(\mathbf{x}) - \langle \mathbf{r}^{(k)}(\mathbf{x}), \mathbf{f}(\mathbf{X}_n) \rangle}{\|\mathbf{r}^{(k)}\|} \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Proof. Write

$$\widehat{\mathbf{f}}(x) - \mathbf{r}^{(k)\top} \mathbf{f}(\mathbf{X}) = \mathbf{r}^{(k)\top} \tilde{\epsilon}_n.$$

To obtain a CLT we check the Lindeberg-Feller condition of $\mathbf{r}^{(k)\top} \tilde{\epsilon}_n$, i.e. for any $\delta > 0$,

$$\lim_n \frac{1}{\|\mathbf{r}^{(k)}\|^2 \sigma^2} \sum_{i=1}^n \mathbb{E} \left[(\mathbf{r}_i^{(k)} \epsilon_i)^2 \mathbf{1}(|\mathbf{r}_i^{(k)} \epsilon_i| > \delta \|\mathbf{r}^{(k)}\| \sigma) \right] = 0.$$

Since $\|\mathbf{k}^{(k)}\|_\infty = O\left(n^{-\frac{1}{d+2}}\right)$ and $\mathbf{P}$ has bounded row sums, we have

$$\|\mathbf{r}^{(k)}\|_\infty \leq \|\mathbf{k}^{(k)}\|_\infty \cdot \|\mathbf{P}\|_1 = O\left(n^{-\frac{1}{d+1}}\right).$$

Furthermore, since $\|\mathbf{r}^{(k)}\| = \Theta(n^{-\frac{1}{2} \frac{1}{d+1}})$, we get

$$\frac{\|\mathbf{r}^{(k)}\|_\infty}{\|\mathbf{r}^{(k)}\|} = O\left(n^{-\frac{1}{2} \frac{1}{d+1}}\right),$$

This allows us to check out the Lindeberg-Feller conditions:

$$\begin{aligned} \sum_{i=1}^n \mathbb{E} \left[(\mathbf{r}_i^{(k)} \epsilon_i)^2 \mathbf{1}(|\mathbf{r}_i^{(k)} \epsilon_i| > \delta \|\mathbf{r}^{(k)}\| \sigma) \right] &\leq \sum_{i=1}^n \mathbf{r}_i^{(k)2} \sqrt{\mathbb{E}[\epsilon_i^4] \cdot \mathbb{E}[\mathbf{1}(|\mathbf{r}_i^{(k)} \epsilon_i| > \delta \|\mathbf{r}^{(k)}\| \sigma)]} \\ &\leq \sum_{i=1}^n \mathbf{r}_i^{(k)2} \sqrt{\mathbb{E}[\epsilon_i^4]} \cdot \sqrt{\mathbb{P}\left(|\epsilon_i| \geq \frac{\delta \|\mathbf{r}^{(k)}\| \sigma}{\mathbf{r}_i^{(k)}}\right)} \\ &\leq \sum_{i=1}^n \mathbf{r}_i^{(k)2} \sqrt{\mathbb{E}[\epsilon_i^4]} \sqrt{2 \exp\left(-\frac{1}{2\sigma^2} \cdot \left(\frac{\delta \|\mathbf{r}^{(k)}\| \sigma}{\mathbf{r}_i^{(k)}}\right)^2\right)} \\ &\leq \|\mathbf{r}^{(k)}\|^2 \exp\left(-O(n^{\frac{1}{d+1}})\right) \rightarrow 0 \end{aligned}$$

Since ϵ is sub-Gaussian noise, the concentraion bound holds by definition of ϵ . $\square$

B.2 Bias Control and Asymptotic Unbiasedness

In Section B.1 we established a conditional central limit theorem: conditioning on the training covariates $\mathbf{X}$, the noise contribution $\mathbf{r}^{(k)}(\mathbf{x})^\top \vec{\epsilon}$ is asymptotically normal with variance σ^2 . To extend this to an unconditional CLT for the estimator, it remains to verify that all *bias terms* arising from the random design vanish at rate $o_p(1)$.

Recall the decomposition

$$\hat{f}^{(k)}(\mathbf{x}) - f^{(k)}(x^{(k)}) = \mathbf{r}^{(k)}(\mathbf{x})^\top \beta + (\mathbf{r}^{(k)}(\mathbf{x})^\top f^{(k)}(\mathbf{X}^{(k)}) - f^{(k)}(x^{(k)})) + \mathbf{r}^{(k)}(\mathbf{x})^\top \sum_{a \neq k} f^{(a)}(\mathbf{X}^{(a)}) + \mathbf{r}^{(k)}(\mathbf{x})^\top \vec{\epsilon}.$$

The last term is the conditional CLT contribution. We now show that the three deterministic terms (baseline, same-feature, cross-feature) all converge to zero at suitable rates.

Lemma B.6 (Baseline Bias). *Let $\beta \in \mathbb{R}^n$ denote the baseline vector. Without subsampling, $\mathbf{r}^{(k)}(\mathbf{x})^\top \beta = 0$ for all $k = 1, \dots, p$.*

Proof. By definition,

$$\mathbf{r}^{(k)}(\mathbf{x})^\top = \mathbf{k}^{(k)}(\mathbf{x})^\top (I - \mathbb{E}\mathbf{S}^{(k)})^\dagger (I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top) [I - (I + \mathcal{K})^\dagger \mathcal{K}],$$

where $\mathcal{K} = \sum_{k=1}^p (I - \mathbb{E}\mathbf{S}^{(k)})^\dagger (I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top) \mathbb{E}\mathbf{S}^{(k)}$. Since every structure matrix has row sum 1, the centering operator annihilates $\mathbf{1}$, so $\mathcal{K}\beta = 0$ and likewise $(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top)\beta = 0$. Thus the entire product vanishes. $\square$

The same feature term bias can be killed by building deeper trees as we collect more and more observations from the underlying data generating process. This can be realized by inheriting assumptions on leaf size from [Zhou and Hooker \(2019\)](#), which we present below:

Remark 1. If subsampling is used, it suffices that the row sum of $\mathbf{S}^{(k)}$ converges to 1 at rate $o(1)$ so that $\mathbf{r}^{(k)}(\mathbf{x})^\top \beta = o_p(1)$.

Lemma B.7 (Same-feature Bias). *Let $f^{(k)}$ be L -Lipschitz in its argument, and let $A_n(\mathbf{x})$ denote the (feature- k) leaf containing $\mathbf{x}$. Suppose the same-feature weights are local in the sense that $r_{n,i}^{(k)}(\mathbf{x}) = 0$ whenever $x_i^{(k)} \notin A_n(\mathbf{x})$, and $\|\mathbf{r}^{(k)}(\mathbf{x})\|_1 = O_p(1)$. Then, writing $\rho_{n,k}(\mathbf{x}) := \sup\{|x_i^{(k)} - x^{(k)}| : x_i^{(k)} \in A_n(\mathbf{x})\}$, we have*

$$|\mathbf{r}^{(k)}(\mathbf{x})^\top f^{(k)}(\mathbf{X}^{(k)}) - f^{(k)}(x^{(k)})| \leq L \rho_{n,k}(\mathbf{x}) \|\mathbf{r}^{(k)}(\mathbf{x})\|_1 = O_p(d_n) = O_p(n^{-1/(d+1)}).$$

In particular, this term is $o_p(1)$.

Proof. By the locality assumption,

$$|\mathbf{r}^{(k)}(\mathbf{x})^\top f^{(k)}(\mathbf{X}^{(k)}) - f^{(k)}(x^{(k)})| = \left| \sum_i r_{n,i}^{(k)}(\mathbf{x}) (f^{(k)}(x_i^{(k)}) - f^{(k)}(x^{(k)})) \right| \leq L \sum_{i: x_i^{(k)} \in A_n(\mathbf{x})} |r_{n,i}^{(k)}(\mathbf{x})| |x_i^{(k)} - x^{(k)}|.$$

The inner distances are bounded by $\rho_{n,k}(\mathbf{x}) \leq \text{diam}(A_n(\mathbf{x}))$, so

$$|\mathbf{r}^{(k)}(\mathbf{x})^\top f^{(k)}(\mathbf{X}^{(k)}) - f^{(k)}(x^{(k)})| \leq L \rho_{n,k}(\mathbf{x}) \sum_i |r_{n,i}^{(k)}(\mathbf{x})| = L \rho_{n,k}(\mathbf{x}) \|\mathbf{r}^{(k)}(\mathbf{x})\|_1.$$

And since $\|\mathbf{r}^{(k)}(\mathbf{x})\|_1 = O_p(1)$, hence the product is $O_p(d_n) = O_p(n^{-1/(d+1)}) = o_p(1)$. $\square$

Thus the same-feature contribution is negligible whenever leaves shrink and the weight vector has bounded ℓ_1 norm.

Lemma B.8 (Rate of Leakage). *Under specified assumptions in section 3, the rate of the leakage term is*

$$|\mathbf{r}^{(k)}(\mathbf{x})^\top \sum_{a \neq k} \mathbf{f}^{(a)}(\mathbf{X})| = O(n^{-\frac{1}{d+1}} \log n) + O\left(\frac{1}{n}\right) = o(1)$$

Proof. We will make use of the radius factor $l_n = \lceil c_1 \log n \rceil$. The leakage term admits the decomposition

$$\mathbf{r}^{(k)}(\mathbf{x})^\top \sum_{a \neq k} \mathbf{f}^{(a)}(\mathbf{X}) = \mathbf{r}^{(k)}(\mathbf{x})|_{D_n}^\top \sum_{a \neq k} \mathbf{f}^{(a)}(\mathbf{X}) + \mathbf{r}^{(k)}(\mathbf{x})|_{D_n^c}^\top \sum_{a \neq k} \mathbf{f}^{(a)}(\mathbf{X})$$

Without loss of generality, we control the behavior of a single dimension a . First bound the near-by term $\mathbf{r}^{(k)}(\mathbf{x})|_{D_n}^\top \mathbf{f}^{(a)}(\mathbf{X})$. By assertion, the underlying additive terms are L -Lipschitz, which guarantees deterministically the rate

$$|\mathbf{r}^{(k)}(\mathbf{x})|_{D_n}^\top \mathbf{f}^{(a)}(\mathbf{X})| \leq L \cdot l_n d_n \left\| \mathbf{r}^{(k)}(\mathbf{x})|_{D_n} \right\|_1 = O(d_n \log n) = O(n^{-\frac{1}{d+1}} \log n) = o(1)$$

Next, we show that the peripheral contribution also goes to 0. Recall by definition

$$|\mathbf{r}^{(k)}(\mathbf{x})|_{D_n^c}^\top = \mathbf{k}^{(k)}(\mathbf{x})|_{D_n^c}^\top (\mathbf{I} - \mathbb{E}\mathbf{S}^{(k)})^{-1} (\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top) [\mathbf{I} - (\mathbf{I} + \mathcal{K})^{-1} \mathcal{K}] f^{(a)}(\mathbf{X}^{(a)})$$

By B.2, we know that $\|\mathbf{I} - \mathbf{K}^{(k)}\|_{\text{op}} \leq \frac{1}{1-\rho_k}$ and $\|\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top\|_{\text{op}} \leq 1$. Hence the goal is to control the rate of the last linear transformation.

Working on the centered subspace $H_0 = \{\mathbf{v} : \mathbf{v}^\top \mathbf{1} \neq 0\}$. We've shown that on the subspace H_0 it holds true that $[\mathbf{I} - (\mathbf{I} + \mathcal{K})^{-1} \mathcal{K}] = (\mathbf{I} + \mathcal{K})^{-1}$. Combining the bound above, we consider the Neumann expansion of the kernel

$$\begin{aligned}
 \left\| \mathbf{r}^{(k)} \right\|_{D_n^c} &= \sum_{|\mathbf{x} - \mathbf{x}_i| > l_n \cdot d_n} |\mathbf{r}_{ni}^{(k)}| \\
 &= \sum_{|\mathbf{x} - \mathbf{x}_i| > l_n \cdot d_n} \left| \sum_j \mathbf{k}_{nj}^{(k)} \left[(\mathbf{I} - \mathbb{E} \mathbf{S}^{(k)})^{-1} (\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^\top) [\mathbf{I} - (\mathbf{I} + \mathcal{K})^{-1} \mathcal{K}] \right]_{j,i} \right| \\
 &\leq C \times \sum_{|\mathbf{x} - \mathbf{x}_i| > l_n \cdot d_n} \left| \sum_{|\mathbf{x} - \mathbf{x}_j| \leq d_n} \mathbf{k}_{nj}^{(k)} \left[[\mathbf{I} - (\mathbf{I} + \mathcal{K})^{-1} \mathcal{K}] \right]_{j,i} \right| \\
 &= C \times \sum_{|\mathbf{x} - \mathbf{x}_j| > l_n \cdot d_n} \left| \sum_{|\mathbf{x} - \mathbf{x}_j| \leq d_n} \mathbf{k}_{nj}^{(k)} \left[(\mathbf{I} + \mathcal{K})^{-1} \right]_{j,i} \right| \\
 &\leq C \times \sum_{|\mathbf{x} - \mathbf{x}_j| > l_n \cdot d_n} \mathbf{k}_{nj}^{(k)} \sum_{|\mathbf{x} - \mathbf{x}_i| > l_n \cdot d_n} \left| \left[(\mathbf{I} + \mathcal{K})^{-1} \right]_{j,i} \right| \\
 &\leq C \times \sum_{|\mathbf{x} - \mathbf{x}_j| > l_n \cdot d_n} \mathbf{k}_{nj}^{(k)} \sum_{|\mathbf{x}_i - \mathbf{x}_j| > (l_n - 1) \cdot d_n} \left| \left[(\mathbf{I} + \mathcal{K})^{-1} \right]_{j,i} \right|
 \end{aligned}$$

By Lemma B.4, we know that $\mathbf{I} + \mathcal{K}$ is positive semidefinite, has bounded inverse and has bounded spectrum from $[1 + \lambda_{\min}(\mathcal{K}), 1 + \lambda_{\max}(\mathcal{K})]$. Hence by proposition C.1, the bound above proceeds as

$$\begin{aligned}
 \left\| \mathbf{r}^{(k)}(\mathbf{x}) \right\|_{D_n^c} &\leq C \sum_{|x_j^{(k)} - x^{(k)}| \leq d_n} k_{nj}^{(k)}(\mathbf{x}) \sum_{|x_i^{(k)} - x^{(k)}| > \ell_n d_n} C_0 q^{\ell_n - 1} \\
 &\leq O(n q^{\ell_n - 1}) \\
 &= O(q^{-1} \cdot n \cdot q^{c_1 \log n}) \\
 &= O(q^{-1} n^{1 - c_1 \log q})
 \end{aligned}$$

where $q = \frac{\sqrt{(1 + \lambda_{\max}(\mathcal{K})) / ((1 + \lambda_{\min}(\mathcal{K})) - 1)}}{\sqrt{(1 + \lambda_{\max}(\mathcal{K})) / ((1 + \lambda_{\min}(\mathcal{K})) + 1)}}$. Choose $c_1 = \frac{2}{\log q}$ we have this contribution being $O(\frac{1}{n}) = o(1)$. And since $\mathbf{f}^{(k)}(\mathbf{X}^{(k)})$ are bounded for all k below M , hence the peripheral term is also of rate $o(1)$. Adding the inner-rim term preserves the rate $o(1)$. $\square$

Collecting Lemmas B.6–B.8 and the we conclude that the random-design bias terms vanish in probability. Together with the conditional CLT for the noise term, Slutsky's theorem implies the full estimator is asymptotically normal (formalized in the next subsection).

B.3 Unconditional Asymptotic Normality

In Section B.2, we have shown that all bias terms vanish to 0 deterministically, given the rate on $\|\mathbf{k}^{(k)}\|$ and $\|\mathbf{r}^{(k)}\|$ hold. However the rate does not come for free. To ensure such assumptions hold, we require our tree structure space to be restricted to support such rate. Formally, we introduce the lemma below.

Lemma B.9. *For given $x \in [0, 1]^d$, suppose we have random sample $(x_1, y_1), \dots, (x_n, y_n)$ for each n . If we restrict the cardinality of all univariate tree space $Q_n^{(k)}$ using any small $\alpha > 0$ s.t.*

$$|Q_n^{(k)}| = O\left(\frac{1}{n} \exp\left(\frac{1}{2} n^{\frac{1}{d+2} - \nu} - n^\alpha\right)\right),$$

then

$$\frac{\hat{\mathbf{f}}^{(k)}(\mathbf{x}) - \mathbf{r}^{(k)}(\mathbf{x})^\top \mathbf{f}(\mathbf{X})}{\|\mathbf{r}^{(k)}(\mathbf{x})\|} \xrightarrow{d} \mathcal{N}(0, \sigma_\epsilon^2).$$

Proof. The proof is similar to Zhou and Hooker (2019) and we direct the audience to the Lemma 4.3. The idea is that we’ve already shown in section B.1 that under the stated assumptions of $|B_n|$ and the growing minimal number of points in tree, we will get the conditional clt for free. Hence the strategy is to show that those two assumptions are met almost surely. Here we simply replace the uniform tree space for the specialized tree space $Q_n^{(k)}, k = 1, \dots, p$.

The sketch of proof is as follow. To show (i), we bound the tail probability of the cardinality of B_n being too large, i.e. $\mathbb{P}(|B_n| > a_n)$ for some growing a_n . This tail probability is bounded by multiplicative Chernoff bound and absolutely summable, which is then a.s. implied by Borel-Cantelli.

As for (ii), we use the similar strategy. Notice that the probability of interest is

$$\mathbb{P}\left(\exists A \in \Pi \in Q_n^{(k)} \text{ s.t. } \inf_{A \in \Pi \in Q_n^{(k)}} \sum_{i=1}^n \mathbf{I}(\mathbf{x}_{i,n} \in A) > Cn^{\frac{1}{d+2}}\right)$$

for some C . The cardinality of $Q_n^{(k)}$ comes in here to serve as a union bound. It can similarly be shown that this probability is also summable and hence (ii) holds a.s.. □

Combining all results from section B.2, B.1, B.3.

Theorem B.2. *Under assumptions on tree space, and all assumptions from section 3, we have*

$$\frac{\hat{\mathbf{f}}^{(k)}(\mathbf{x}) - \mathbf{f}^{(k)}(\mathbf{x})}{\|\mathbf{r}^{(k)}(\mathbf{x})\|} \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

for all $k = 1, \dots, p$.

Proof. We break the numerator into three terms.

$$\frac{\hat{\mathbf{f}}^{(k)}(\mathbf{x}) - \mathbf{f}^{(k)}(\mathbf{x})}{\|\mathbf{r}^{(k)}(\mathbf{x})\|} = \frac{1}{\|\mathbf{r}^{(k)}(\mathbf{x})\|} (\mathbf{r}^{(k)}(\mathbf{x})^\top \beta + (\mathbf{r}^{(k)}(\mathbf{x})^\top f^{(k)}(\mathbf{X}^{(k)}) - f^{(k)}(x^{(k)})) + \mathbf{r}^{(k)}(\mathbf{x})^\top \sum_{a \neq k} f^{(a)}(\mathbf{X}^{(a)}) + \mathbf{r}^{(k)}(\mathbf{x})^\top \tilde{\epsilon})$$

Under restricted tree space assumptions, all terms except the last term are $o_p(1)$, and the last term is conditionally asymptotically normal. By Slutsky’s theorem, the sum has a limiting distribution being $\mathcal{N}(0, \sigma^2)$. □

The CLT holds trivially for the baseline vector as well, by definition of the fixed point it converge to.

Theorem B.3 (CLT for Baseline Vector). *Let $\hat{\beta} = \frac{1}{n} \sum_{i=1}^n y_i$ denote the baseline estimator returned by Algorithm 1. Under Assumption 3, the random vector $\sqrt{n}(\hat{\beta} - \beta)$ converges in distribution to a Gaussian:*

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

where $\beta = \mathbb{E}[Y]$ is the population intercept.

Proof. By construction of the algorithm, the intercept is always updated by the global mean of residuals. At convergence this coincides with the empirical mean of the responses:

$$\hat{\beta} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i.$$

Since $\mathbf{y}_i = f(\mathbf{x}_i) + \epsilon_i$ with $\mathbb{E}[\epsilon_i] = 0$ and $\text{Var}(\epsilon_i) = \sigma^2 < \infty$, we have $\mathbb{E}[\mathbf{y}_i] = \beta$. Because the $\{\mathbf{y}_i\}$ are i.i.d. sub-Gaussian with finite variance, the Lindeberg–Feller central limit theorem applies, yielding

$$\sqrt{n}(\hat{\beta} - \beta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (y_i - \mathbb{E}[y_i]) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

□

C Stochastic Contraction Mapping Theorem

Theorem C.1. *Given $\mathbb{R}^d$ -valued stochastic process $\{\mathbf{z}_t\}_{t \in \mathbb{N}}$, a sequence of $0 < \lambda_t \leq 1$, define*

$$\begin{aligned}\mathcal{F}_0 &= \emptyset, \mathcal{F}_t = \sigma(\mathbf{z}_1, \dots, \mathbf{z}_t), \\ \epsilon_t &= \mathbf{z}_t - \mathbb{E}[\mathbf{z}_t | \mathcal{F}_{t-1}].\end{aligned}$$

We call $\mathbf{z}_t$ a stochastic contraction if the following properties hold

1. *Vanishing coefficients*

$$\sum_{t=1}^{\infty} (1 - \lambda_t) = \infty, \text{ i.e. } \prod_{t=1}^{\infty} \lambda_t = 0.$$

2. *Mean contraction*

$$\|\mathbb{E}[\mathbf{z}_t | \mathcal{F}_{t-1}]\| \leq \lambda_t \|\mathbf{z}_{t-1}\|, \text{ a.s..}$$

3. *Bounded deviation*

$$\sup \|\epsilon_t\| \rightarrow 0, \quad \sum_{t=1}^{\infty} \mathbb{E}[\|\epsilon_t\|^2] < \infty.$$

In particular, a multidimensional stochastic contraction exhibits the following behavior

1. *Contraction*

$$\mathbf{z}_t \xrightarrow{\text{a.s.}} 0.$$

2. *Kolmogorov inequality*

$$P\left(\sup_{t \geq T} \|\mathbf{z}_t\| \leq \|\mathbf{z}_T\| + \delta\right) \geq 1 - \frac{4\sqrt{d} \sum_{t=T+1}^{\infty} \mathbb{E}[\epsilon_t^2]}{\min\{\delta^2, \beta^2\}} \quad (3)$$

holds for all $T, \delta > 0$ s.t. $\beta = \|\mathbf{z}_T\| + \delta - \sqrt{d} \sup_{t > T} \|\epsilon_t\| > 0$.

Proposition C.1 (Demko–Moss–Smith, General Sparse Exponential Decay). *Let A be positive definite, bounded, and boundedly invertible on $\ell^2(S)$. For any such A , define the support sets*

$$S_n(A) := \bigcup_{k=0}^n \{(i, j) : (A^k)_{ij} \neq 0\}, \quad D_n(A) := (S \times S) \setminus S_n(A).$$

Then there exist constants $C_0 < \infty$ and $q \in (0, 1)$, depending only on the spectral bounds of A , such that

$$\sup\{|A^{-1}(i, j)| : (i, j) \in D_n(A)\} \leq C_0 q^{n+1}.$$

Here the decay factor q is given explicitly by

$$q = \frac{\sqrt{r} - 1}{\sqrt{r} + 1}, \quad r = \frac{b}{a},$$

where $[a, b]$ is the smallest interval containing the spectrum of A .